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Targeting Airway Remodeling in Severe Asthma: Is There a Window of Opportunity for Biologic Therapy Predicting Effects Using Causal Artificial Intelligence?

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ABSTRACT

Airway remodeling is increasingly recognized as a major determinant of asthma progression, fixed airflow limitation, and long-term morbidity, particularly in severe disease. Although biologic therapies have transformed outcomes by reducing exacerbations and systemic corticosteroid exposure, their potential to modify structural airway trajectories—and whether a time-sensitive "window of opportunity" exists—remains uncertain. Here, we provide a progressive landscape integrating mechanistic remodeling pathways with measurable structural readouts (biopsy-derived indices and quantitative imaging) and emerging digital biomarkers derived from connected respiratory technologies. We propose an operational framework linking mechanism → biomarker → remodeling readout → timing decision, and we outline a 1-, 3-, and 5-year research roadmap in which advanced artificial intelligence (AI) methods (multimodal learning, causal inference, federated learning, and digital-twin architectures) evolve in parallel with wearable and smart-inhaler ecosystems. This landscape aims to standardize endpoints, sharpen trial design, and accelerate a shift from symptom control toward credible disease modification in severe asthma.

1 | Introduction: Why "Timing" Is Now the Central Question

Severe asthma remains a major global health concern and is more prevalent in the adult population; however, children and adolescents are at higher risk of severe exacerbations, with significant impacts on quality of life, increased exposure to oral corticosteroids (OCS), and high healthcare utilization. The introduction of biologic therapies has markedly changed the clinical approach to this disease, providing precision-based strategies that directly target immunological drivers of disease. Even in well-structured pathways, many patients require biologics to reduce exacerbations and steroid toxicity, reshaping the goal from controlling acute events to altering the longer-term trajectory of disease [1, 2].

Central to this trajectory is airway remodeling, a complex process encompassing epithelial injury and barrier dysfunction, subepithelial fibrosis, extracellular matrix (ECM) reorganization, increased airway smooth muscle mass (ASM), glandular alterations, angiogenesis, altered innervation, and mucus pathology. These structural changes contribute to airway hyperresponsiveness, progressive lung-function decline, and, in a subset of patients, the development of fixed airflow limitation [3–6].

Sebastiano Gangemi and Sara Manti contributed equally to this work.

Crucially, airway remodeling should not be viewed solely as a downstream consequence of persistent inflammation; rather, structural cells function as active immunological players, and sustained epithelial–mesenchymal signaling together with mechanotransduction pathways maintain remodeling processes even in the absence of ongoing inflammatory activity [3–7].

Within this framework, the concept of a therapeutic "window of opportunity" is biologically plausible. Introducing biologics early in the disease course offers a crucial therapeutic advantage. Acting during this window allows the therapy to effectively impact the trajectory of tissue damage, preserving the underlying architecture of the lungs before irreversible structural changes set in. Conversely, if biologic therapy is delayed, the airway remodeling becomes permanent, causing the airflow obstruction to become "fixed" and irreversible. While initiating a biologic at this late stage remains beneficial for lowering the frequency of acute exacerbations, it can no longer restore lost lung function or eliminate baseline symptoms [6–8]. This hypothesis has gained renewed relevance in light of population-based and severe-asthma cohort data demonstrating that exacerbations are associated with accelerated lung-function decline, underscoring the clinical importance of preventing recurrent inflammatory insults that may drive structural progression [9, 10]. Moreover, longitudinal analyses in severe asthma reveal heterogeneous inflammatory patterns in relation to lung-function decline over time, reinforcing the need to move beyond static biomarkers toward a trajectory-based conceptualization of disease evolution [11]. Two converging advances now make the hypothesis testable. First, quantitative imaging has matured into a scalable platform for remodeling-relevant endpoints: chest tomography (CT) enables objective mucus plug scoring and airway metrics; functional imaging approaches, including hyperpolarized-gas magnetic resonance imaging (MRI), capture persistent ventilation defects and heterogeneity that relate to clinically meaningful disease burden [12–17]. Second, connected respiratory technologies—smart inhalers, remote monitoring, wearables, smartphone cough sensing, and exposure tracking—enable continuous real-world signals to define temporal "risk states," objectively characterize adherence and rescue dependence, and support testable timing strategies [18–25]. These tools also enable modern analytics, including multimodal learning for dynamic phenotyping and causal inference approaches that estimate counterfactual timing effects—namely, assessing how outcomes might differ if the timing of an intervention or exposure were shifted earlier or later than what occurred.

This landscape brings together (i) the mechanistic basis and heterogeneity of airway remodeling, (ii) complementary measurement approaches—including biopsy-derived indices, advanced imaging, and digital and breath-based biomarkers, (iii) emerging evidence that biologic therapies may modulate remodeling-related outcomes, and (iv) a forward-looking research roadmap in which connected device ecosystems and artificial intelligence (AI) co-evolve across 1-, 3-, and 5-year horizons. Building on these elements, we propose a reusable operational framework—linking mechanisms to biomarkers, remodeling readout, and timing decisions—to accelerate the transition toward credible disease-modification strategies in severe asthma (Figure 1).

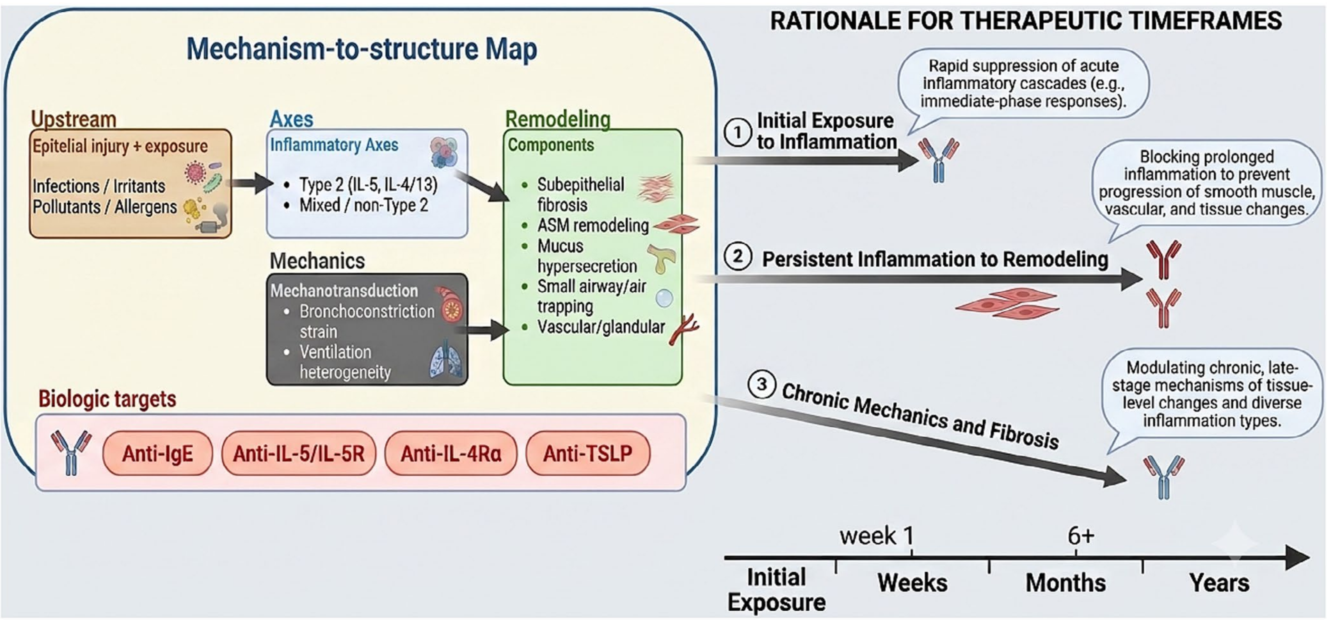

FIGURE 1 | Mechanism-to-structure map (biologic targets and remodeling components). Conceptual framework linking upstream triggers (epithelial injury, infections/exposures, and epithelial alarmins) to downstream inflammatory axes (type 2 and mixed/non–type 2 contributions) and mechanotransduction (bronchoconstriction-driven strain and ventilation heterogeneity), converging on key airway remodeling components. Remodeling domains include subepithelial fibrosis (reticular basement membrane thickening and extracellular matrix (ECM) remodeling), airway smooth muscle hypertrophy/hyperplasia, goblet cell metaplasia and mucus hypersecretion, small airway dysfunction with air trapping, and vascular/glandular remodeling. Biologic intervention points (anti-IgE, anti–IL-5/IL-5R, anti–IL-4Rα, and anti-TSLP) are positioned along the mechanistic cascade to illustrate how pathway-specific suppression of inflammation and upstream signals could plausibly yield time-sensitive effects on the structural trajectory, supporting the "window-of-opportunity" hypothesis.

2 | Remodeling in Severe Asthma: Mechanisms, Compartments, and Heterogeneity

2.1 | Remodeling Is Structural Biology, Not "Chronic Inflammation" Alone

Several studies consistently demonstrated that airway remodeling in asthma is a multicompartmental process driven by coordinated heterogeneous "structural programs," including epithelial fragility and impaired repair, ECM remodeling with subepithelial fibrosis, ASM hypertrophy and hyperplasia, as well as microvascular and glandular alterations that collectively determine airway mechanics and hyperresponsiveness, all of which may be differentially expressed or hold varying relevance at the individual level. [3–6]. Early morphometric and biopsy-based investigations, complemented by more recent imaging and systems-level analyses, have established that these changes can start precociously in disease progression and maintain independently across airway layer compartments.

Within this framework, the airway epithelium occupies a central sentinel position, integrating environmental injury, allergen exposure, and immune activation with repair and differentiation programs that, when chronically engaged, may become maladaptive and promote sustained remodeling, reinforcing the concept of the epithelium as an active driver rather than a passive target of disease [6, 26–28].

Importantly, mechanical forces have emerged as independent contributors to remodeling. Experimental and clinical studies demonstrated that repeated bronchoconstriction itself can induce structural changes—including increased ASM mass and matrix remodeling—even in the absence of overt inflammation, supporting mechano-transduction as a parallel remodeling pathway and providing a plausible biological explanation for inflammation–structure decoupling observed in some patients with severe asthma [7, 29–31].

Immune–structural interactions further reinforce this paradigm. The preferential infiltration of mast cells within ASM bundles exemplifies compartmentalized immune activity in remodeling-relevant tissue and has been linked to persistent airway dysfunction, bronchial hyperresponsiveness, and fixed airflow limitation despite improvements in systemic or sputum-based inflammatory biomarkers [22, 32, 33]. More recent integrative syntheses position ASM remodeling as a key determinant of clinical and biological heterogeneity in asthma, highlighting it as both a marker of disease chronicity and a potential target for structural intervention [23, 34, 35]. Collectively, these insights underscore why remodeling-sensitive endpoints—beyond symptom control and exacerbation reduction—are essential for substantiating claims of true disease modification.

2.2 | Small Airways and Mucus Plugging as Tractable Remodeling-Associated Phenotypes

Mucus plugging has emerged as a consequential structural–functional phenomenon in severe asthma, reflecting more than a transient manifestation of airway inflammation. CT studies have consistently demonstrated a strong association between the presence and extent of mucus plugs and key markers of disease severity, including eosinophilic airway inflammation and fixed airflow obstruction (FAO). In particular, higher mucus plug scores correlate with reduced forced expiratory volume in 1 Second (FEV_1) and increased blood and sputum eosinophilia, supporting the concept that mucus accumulation is tightly linked to type 2 inflammatory pathways and downstream functional impairment [13].

Importantly, longitudinal imaging data indicate that mucus plugs are not merely episodic findings related to acute exacerbations. Plugs often persist over time, and changes follow closely changes in lung function, even in clinically stable patients. This temporal stability suggests that mucus plugging represents a durable structural–functional trait rather than a fluctuating symptom correlate, reinforcing its relevance as a core disease feature in severe asthma [14].

Moreover, advanced imaging has further linked mucus plugging to local pulmonary abnormalities in ventilation. CT and hyperpolarized-gas MRI reveal that mucus plugs colocalize with persistent ventilation defects, providing a mechanistic link between luminal airway obstruction and spatially heterogeneous ventilation [15, 16, 36].

Taken together, this body of evidence positions mucus burden—and critically, its spatial distribution—as an attractive class of imaging endpoints for timing and intervention studies. Mucus plugs are quantifiable at scale using standard CT, clinically meaningful in relation to symptoms and lung function, spatially resolved, and plausibly responsive to targeted therapies. As such, they offer a compelling opportunity to link structural pathology with functional outcomes and treatment effects in severe asthma.

2.3 | Treatable Traits and Remission: Reframing the Timing Hypothesis

Treatable-trait paradigms propose moving beyond static diagnostic categories by decomposing chronic airway disease into modifiable biological, behavioral, and environmental domains. This enables dynamic, patient-centered management rather than rigid phenotyping, which often fails to capture disease evolution [28, 29]. However, growing evidence suggests that approach may not consistently address disease complexity. Severe asthma is thus viewed as a variable expression of multiple traits that evolve over time and respond differently to therapy [30]. Consequently, treatment impact depends not only on efficacy but also on the disease phase at initiation.

Timing becomes central: biologic benefits may vary across phases of instability, structural progression, or established FAO. Identifying these phases through dynamic biomarkers and repeated monitoring is essential to optimize outcomes.

Clinical remission, although increasingly achievable, may not fully reflect disease status. Consensus definitions include symptom control, exacerbation freedom, and lung function stability [31], while real-world studies identify predictors of remission after biologics [32, 33, 37]. However, remission rarely incorporates structural endpoints. Integrating imaging and digital biomarkers could redefine remission as a change in disease trajectory and provide a measurable framework to test the window of opportunity hypothesis.

3 | Measuring Remodeling Longitudinally: Biopsies, Imaging, Digital and Breath-Based Biomarkers

3.1 | Tissue Anchors: What Biopsies Validate—and What They Cannot Scale

Bronchial biopsy allows the direct quantification of key structural features such as subepithelial fibrosis, ECM organization, and ASM remodeling, which are not reliably reflected by clinical or functional measures [3–5]. Accordingly, in disease-modification research, biopsy remains essential to validate and calibrate imaging-derived and other noninvasive surrogate endpoints, ensuring that observed changes reflect true structural modification rather than transient physiological variability. However, due to its invasive nature and sampling limitations, its use is restricted for repeated or population-scale longitudinal assessment. Also, spatial heterogeneity of airway pathology further limits the generalizability of single-site samples, especially when the research objective is to characterize whole-lung disease dynamics over time.

3.2 | Quantitative Imaging as Structural-Functional Surveillance

Quantitative pulmonary imaging provides repeatable organ-level surrogates for remodeling and small airway disease. CT captures airway geometry, wall thickness, luminal caliber, air trapping, and mucus plug burden; functional imaging approaches capture ventilation heterogeneity and persistent ventilation defects relevant to clinical burden [12–17]. Imaging supports timing hypotheses by (i) quantifying baseline structural burden to stratify patients and (ii) generating longitudinal trajectories to test whether early biologic initiation alters the slope or stabilizes structural–functional endpoints [13–15].

3.3 | Connected Technologies and Digital Biomarkers: Capturing the "Time Axis"

Timing hypotheses require temporal granularity. Digital inhalers and remote monitoring provide objective adherence and rescue-use data, distinguishing poor adherence from refractory biology and thereby preventing misclassification in biologic eligibility [18, 22]. Randomized evidence shows that electronic monitoring with reminders can improve adherence and outcomes, even if only over short periods of time [19]. Protocol-driven initiatives aim to identify early digital markers of attacks using sensors and AI, integrating physiology, behavior, and the environment [20]. Smartphone-based cough counting has been tested in acute exacerbations, supporting objective symptom capture in real-world settings [23]. Beyond feasibility, smart-inhaler programs are now being evaluated in pragmatic cluster-randomized designs in primary care, explicitly targeting adherence and outcomes over longer follow-up [24]. Additionally, randomized clinical trial (RCT) evidence shows that digital monitoring through smart spacer-driven education can reduce inhaler errors, a crucial upstream determinant of effective therapy exposure [25]; (Figure 2). Collectively, these developments position digital monitoring technologies as essential enablers of time-sensitive, precision approaches to asthma management, with direct relevance for testing timing-dependent treatment hypotheses.

3.4 | Breathomics/eNose as Emerging Non-Invasive Biology Aligned to Treatable Traits and AI

Breathomics—analysis of volatile organic compounds (VOCs) using mass spectrometry or sensor-based electronic noses—offers a noninvasive window into airway and systemic metabolism potentially linked to inflammation and disease activity. Because VOCs arise from complex interactions between host metabolism, airway inflammation, oxidative stress, and the microbiome, breathomics has the potential to capture biological processes that are not readily accessible through conventional blood or sputum biomarkers, while remaining well suited for repeated and real-world sampling. A literature review describes breathomics applications in asthma and chronic obstructive pulmonary disease (COPD) for phenotyping, exacerbation prediction, and treatment stratification [39]. In asthma, VOC profiles have been associated with inflammatory endotypes, symptom burden, and treatment response, suggesting that breathomics may function as an integrative biomarker reflecting both airway-specific and systemic processes [40, 41].

Established work on eNose breathprints frames key methodological issues and questions for translating breath profiling into reliable clinical biomarkers [42]. Breathomics has been explicitly positioned as complementary to treatable traits, emphasizing the need for standardized populations, sampling methods, and validation strategies [43]. Technology-focused reviews summarize respiratory applications of electronic nose systems and their pattern-recognition foundations [44]. More recent asthma-focused reviews synthesize the role of VOCs for diagnosis and monitoring, contrasting GC–MS and eNose approaches and highlighting standardization needs [45]. In the context of timing, breathomics could provide a scalable "near-real-time biology" layer that complements imaging (structure) and digital biomarkers (time) and is naturally amenable to multimodal AI. In the context of timing-dependent hypotheses, breathomics occupies a distinctive and potentially powerful niche. By providing a scalable, repeatable, and near-real-time readout of biological activity, breathomics could support a dynamic "biology layer" that complements imaging-based structural endpoints and digitally derived temporal biomarkers.

4 | Biologics: Robust Clinical Benefit, Emerging Structural Signals, Unresolved Timing Causality

4.1 | What Biologics Clearly Do—and What They Do Not Yet Prove

Biologics reduce exacerbations and OCS exposure in selected severe asthma populations across multiple pathways [35, 46–62]. Accordingly, beyond direct airway damage,

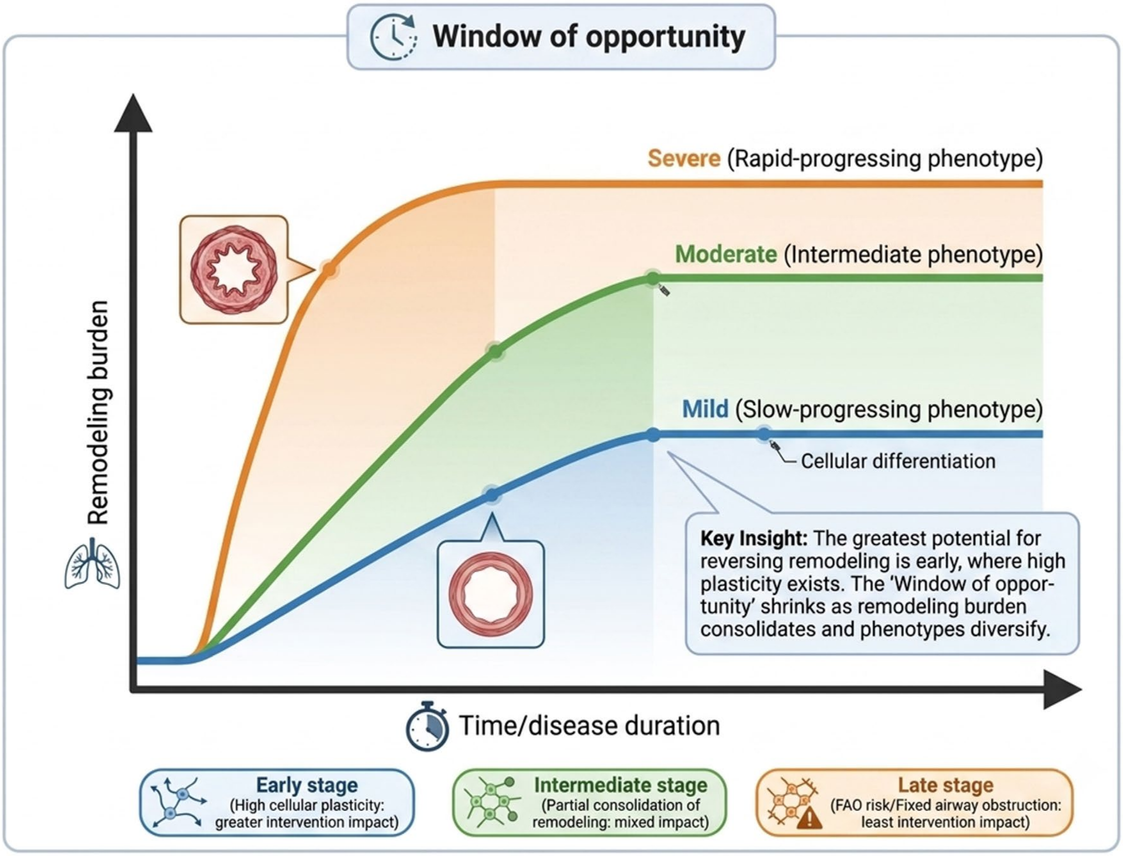

FIGURE 2 | Window of opportunity (structural trajectory model). Conceptual trajectories of remodeling burden over disease duration comparing early biologic initiation versus delayed initiation. In line with the findings reported by Varricchi G et al. (ref. [38]), the model illustrates a higher expected impact of biologics during an early "high-plasticity" phase, progressively attenuating during intermediate stages and approaching limited reversibility in late stages characterized by increased risk of fixed airflow obstruction. Suggested phase-aligned endpoints include early biomarker/digital instability signals and intermediate-to-late imaging-based surrogate trajectories (e.g., CT mucus burden, air trapping, ventilation defects).

delaying targeted therapy heavily increases a patient's reliance on systemic CS. A primary goal of early biologic adoption is mitigating the cumulative steroid burden, shifting patients away from the toxicities of long-term steroid use. Moreover, failing to intervene early drastically alters a patient's long-term health trajectory. Clinical data across many chronic diseases consistently demonstrate a direct correlation between high cumulative corticosteroid use and both poorer temporal trajectories and higher overall mortality. Early biologic intervention is therefore not just a matter of symptom control but a critical strategy to mitigate these fatal risks. Biologics should not be viewed merely as a last resort for end-stage disease, but as a disease-modifying intervention that yields the highest clinical dividend when introduced before airway remodeling becomes irreversible. Overall, the odds of achieving clinical remission, reducing the exacerbation rate and oral corticosteroid use, having better asthma control, and improving lung function post-biologic were most likely in those with reduced time to biologic initiation.

Anti-IgE therapy (omalizumab) demonstrated efficacy in both severe allergic asthma and nonallergic asthma, although it is not licensed for this phenotype [32, 33, 35]. Anti–IL-5 pathway therapies showed exacerbation reduction and steroid-sparing effects [46–51], and anti–IL-5Rα therapy expanded evidence with large trials and extension data [52–54]. Dupilumab improved outcomes in moderate-to-severe uncontrolled asthma and OCS-dependent disease, with biomarker-informed subgroup effects and long-term extension evidence [55–58]. Tezepelumab demonstrated broad efficacy across biomarker strata and has long-term extension data supporting sustained benefit [59–62].

However, pivotal programs were designed primarily around exacerbations and symptoms; structural endpoints were rarely primary outcomes. A window of opportunity is not merely a question of short-term control—it is whether *earlier initiation changes the future structural–functional trajectory.* To address that, endpoints must include remodeling-relevant measures and trajectories.

4.2 | Evidence Synthesis: What Meta-Analyses Imply About Heterogeneity and Staging

Systematic reviews and meta-analyses of anti–IL-5 therapies, including mepolizumab, reslizumab, and benralizumab, consistently show robust reductions in exacerbation frequency and meaningful steroid-sparing effects in patients with severe eosinophilic asthma [63–65]. These benefits are reproducible across trials and real-world studies, firmly establishing IL-5 pathway inhibition as an effective strategy for controlling exacerbation-prone disease driven by eosinophilic inflammation.

In contrast, effects on lung function and patient-reported symptoms change widely across studies. Improvements in FEV1, symptom severity, and quality of life vary substantially in magnitude and consistency, with some trials showing modest gains and others reporting limited or no additional benefit beyond exacerbation frequency. This variability could also be due to differences in baseline structural burden, disease stage, and the degree of coupling between active inflammation and established airway remodeling. In patients with advanced or consolidated structural changes, suppression of eosinophilic inflammation alone may be insufficient to reverse airflow limitation or symptom burden.

Initiation during phases of active inflammatory–structural coupling may allow biologics to attenuate or slow remodeling trajectories, whereas later initiation—after FAO and architectural changes are established—may primarily yield anti-exacerbation benefits without substantial functional recovery. Thus, the timing dependency provides a plausible biological explanation for the observed dissociation between exacerbation outcomes and lung function responses in clinical trials [38].

Similarly, experiences from anti–IL-13 development programs, most notably with lebrikizumab, further assessed the importance of biomarker enrichment and endpoint selection when targeting pathways implicated in airway remodeling. Early trials demonstrated that IL-13 blockade could improve lung function preferentially in biomarker-defined subgroups (e.g., periostin-high patients), highlighting the need to align patient selection with the underlying biology being targeted [66, 67]. Subsequent phase 3 programs underscored how the choice of primary endpoints—symptom control versus lung function or structural measures—can strongly influence perceived efficacy, particularly when the therapeutic hypothesis centers on modification of airway structure rather than short-term symptom relief [68].

4.3 | Tissue-Level Structural Signals: Plausibility That Remodeling Is Modifiable

Mechanistic trials anchored in bronchial tissue show that pathway modulation can influence remodeling biology. Anti–IL-5 reduced ECM protein deposition in the bronchial subepithelial basement membrane [69], and related work demonstrated significant though incomplete depletion of airway tissue eosinophils [70]. Anti–IL-5 also influenced eosinophil progenitor biology, supporting plausible downstream structural implications [71]. For anti-IgE, therapy, omalizumab has been associated with reduced reticular basement membrane thickness and fibronectin deposition [72] and with remodeling-related marker changes in bronchoalveolar lavage [73]. These studies do not prove disease modification, but they establish that structural biology can respond to targeted therapy—an essential prerequisite for the window-of-opportunity science.

4.4 | Imaging Signals: Mucus Burden and Ventilation Defects as Modifiable Targets

Imaging has yielded particularly compelling remodeling-relevant signals. Anti–IL-5Rα therapy has been associated with reductions in CT mucus score and improvements in hyperpolarized-gas MRI ventilation defects over multiyear treatment, with baseline imaging metrics predicting longer-term outcomes [74]. Single-dose benralizumab studies showed early improvements in ventilation defect measures [75], indicating that relief of eosinophil-driven airway obstruction can rapidly translate into improved regional ventilation, even before global lung function changes become apparent [76]. CT-based studies also reported benralizumab effects on mucus plugs in severe eosinophilic asthma [60]. Accordingly, in patients with severe eosinophilic asthma a reduction in mucus plugs has been reported following benralizumab therapy, linking eosinophil depletion to downstream effects on airway luminal patency and mucus clearance [13]. These data position mucus burden and ventilation heterogeneity as tractable endpoints for timing studies and candidate surrogates for structural trajectory modification.

4.5 | Fixed Airflow Limitation: Why Late Benefit May Not Equal Modification

Fixed airflow obstruction (FAO) is the clinical manifestation of cumulative structural and functional airway changes, marking a transition from reversible bronchoconstriction to irreversible alterations. These changes limit full functional recovery even when inflammation is effectively controlled.

Irreversible airflow limitation is associated with disease progression toward less modifiable structural states, driven by long disease duration, recurrent exacerbations, persistent inflammation, smoking, mucus plugging, and early-life lung deficits [8]. Together, these factors promote airway remodeling that stabilizes airflow limitation over time.

Despite this, patients with FAO can still benefit from biologic therapies, showing reductions in exacerbations, improved symptom control, and steroid-sparing effects, particularly in eosinophilic asthma [77]. However, lung function gains are often attenuated, suggesting that while inflammatory drivers remain modifiable, structural damage is relatively fixed [77].

These findings support timing-dependent treatment strategies. Once structural consolidation is established, biologics may stabilize disease without reversing its trajectory. Early identification of patients in high-risk phases—where inflammation and remodeling are still coupled—is therefore essential to modify long-term outcomes and prevent progression to FAO [77].

In this framework, FAO represents a marker of a missed therapeutic opportunity, underscoring the need to integrate longitudinal functional, imaging, and biological data to guide earlier intervention.

5 | Bronchial Thermoplasty as a Structural Comparator

Because it directly targets ASM, bronchial thermoplasty (BT) is particularly informative for disease-modification research. By delivering controlled thermal energy to the airway wall, BT induces a sustained reduction in ASM mass, providing an example of an intervention with a clearly defined structural mechanism of action. Also, BT provides direct evidence that airway architecture is not immutable and that targeted interventions can induce measurable, durable structural change.

Histological and imaging studies have documented significant reductions in ASM area and volume following BT, accompanied by changes in remodeling-related markers within the airway wall that are maintained over time [78–80]. Importantly, structural changes following BT have been associated with improvements in clinical outcomes, such as reductions in exacerbation frequency and emergency department visits, although in the absence of significant improvement in lung function.

Although BT is mechanistically distinct from biologic therapies—acting through physical reduction of ASM rather than modulation of inflammatory pathways—it provides, indirect evidences, that airway structure is modifiable in adults with severe asthma, as well as the importance of aiming to change the disease course rather than short-term symptom control.

6 | The Window-of-Opportunity Problem Is Causal: What AI Must Do (and Avoid)

6.1 | Prediction Is Necessary Infrastructure—But Insufficient

Machine learning can predict short-term exacerbation risk from clinical datasets [81, 82], and reviews summarize the rapidly growing AI/ML ecosystem in asthma [83]. Yet data-driven phenotyping and ML can fail when confounding, limited generalizability, and outcome misalignment are ignored—particularly if models optimize near-term events rather than longer-horizon structural trajectories [84]. Prediction can help define operational "risk windows" and enrich trials, but it cannot by itself answer whether early initiation changes structural outcomes (Figure 2).

6.2 | Causal Inference and Target Trial Emulation: Making Timing Estimable

A window of opportunity is inherently causal: what would happen if a biologic were started earlier rather than later. Target trial emulation provides a practical framework for causal inference using observational data by explicitly specifying eligibility, strategies, outcomes, follow-up, and causal contrasts [85]. Methodological guidance highlights pitfalls in observational analyses and emphasizes careful handling of time-varying confounding—which is central to timing research where

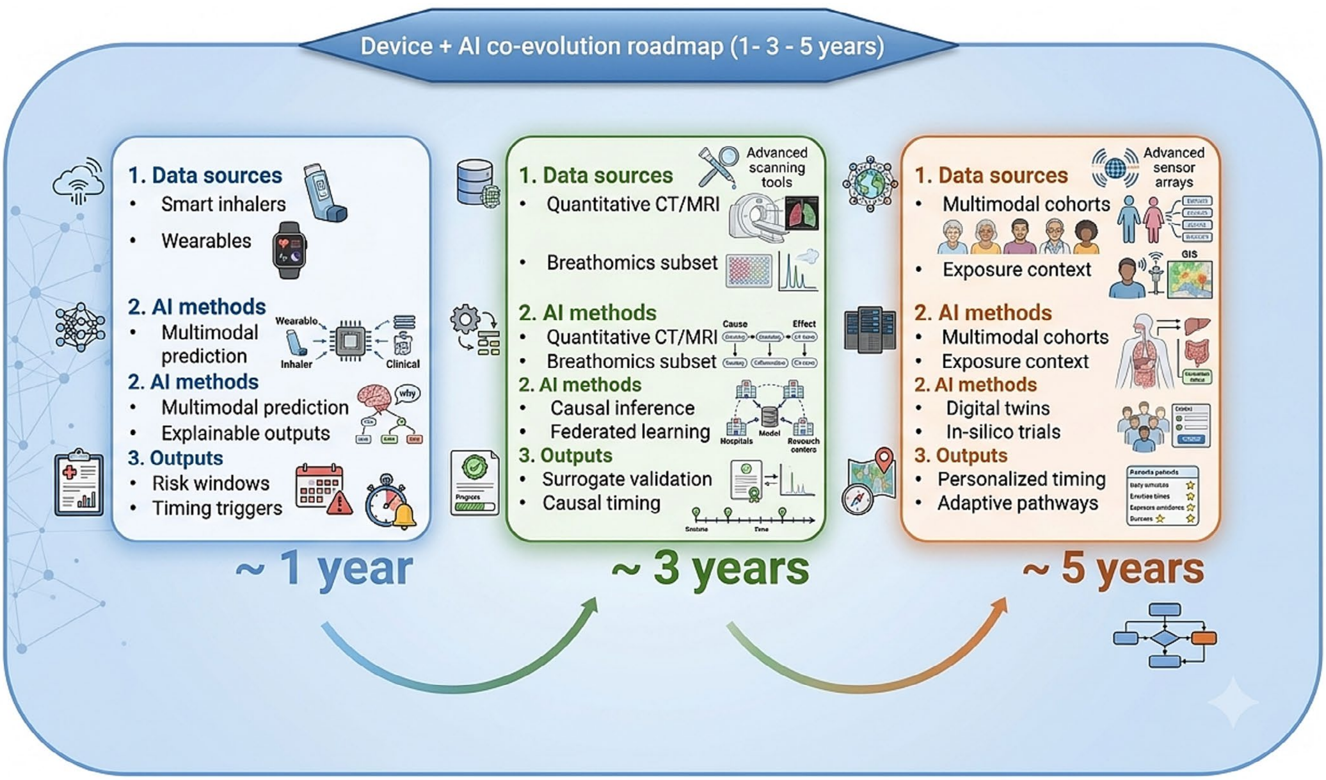

FIGURE 3 | Co-evolution roadmap: Connected devices and AI (1–, 3–, and 5–years). Progressive landscape depicting how maturation of connected respiratory technologies enables increasingly advanced AI approaches. In the near term (~1 year), smart inhalers and wearables support multimodal prediction and explainable identification of high-risk windows and timing triggers. Within ~3 years, standardized quantitative imaging and breathomics subsets enable causal inference (target trial emulation) and privacy-preserving multicenter learning. Within ~5 years, multimodal longitudinal cohorts support digital twins and in silico trial augmentation to optimize individualized timing strategies and disease-modification research.

exacerbations, steroid bursts, adherence, and biomarker dynamics evolve over time [86]. These tools enable estimation of early versus delayed initiation effects on imaging and digital trajectories when pragmatic RCTs are not feasible.

6.3 | Scaling Timing Science: Federated Learning and Privacy-Preserving Ecosystems

Timing research requires large, diverse datasets spanning EHRs, imaging repositories, and device ecosystems. Federated learning and privacy-preserving ecosystems provide a secure framework for multicenter model development by allowing models to be trained locally at each site without transferring raw patient data. Instead, only model updates (e.g., weights or gradients) are shared and aggregated, protecting sensitive information while enabling learning from large, heterogeneous datasets. Privacy-preserving techniques—including encryption, differential privacy, and secure multiparty computation—further ensure that data collected through telemedicine, remote monitoring, or connected devices can be analyzed safely under governance constraints [87, 88]. These infrastructures are particularly relevant for connected device time series, which are highly sensitive and fragmented across platforms, allowing dynamic, patient-centered precision analytics while maintaining privacy and regulatory compliance.

6.4 | Self-Supervised Learning and Digital Twins: A Plausible 5-Year Horizon

Wearables and smart inhalers generate high-frequency time series data with limited labels. Self-supervised contrastive learning can learn robust representations from unlabeled medical time-series data, enabling downstream prediction, clustering, and trajectory modeling [89]. Home monitoring in asthma is increasingly framed as a pathway toward digital twins—patient-specific models updated by sensor streams to simulate trajectories and intervention responses [89, 90]. Digital twin frameworks for chronic lung diseases describe their feasibility, benefits, and challenges, supporting a plausible 5-year horizon, provided validation and governance keep pace [89, 91].

7 | Operational Framework: Mechanism → Biomarker → Remodeling Readout → Timing Decision

To render the window-of-opportunity hypothesis testable in research and clinical practice, a reusable operational framework integrating mechanism, measurement, and timing has been proposed.

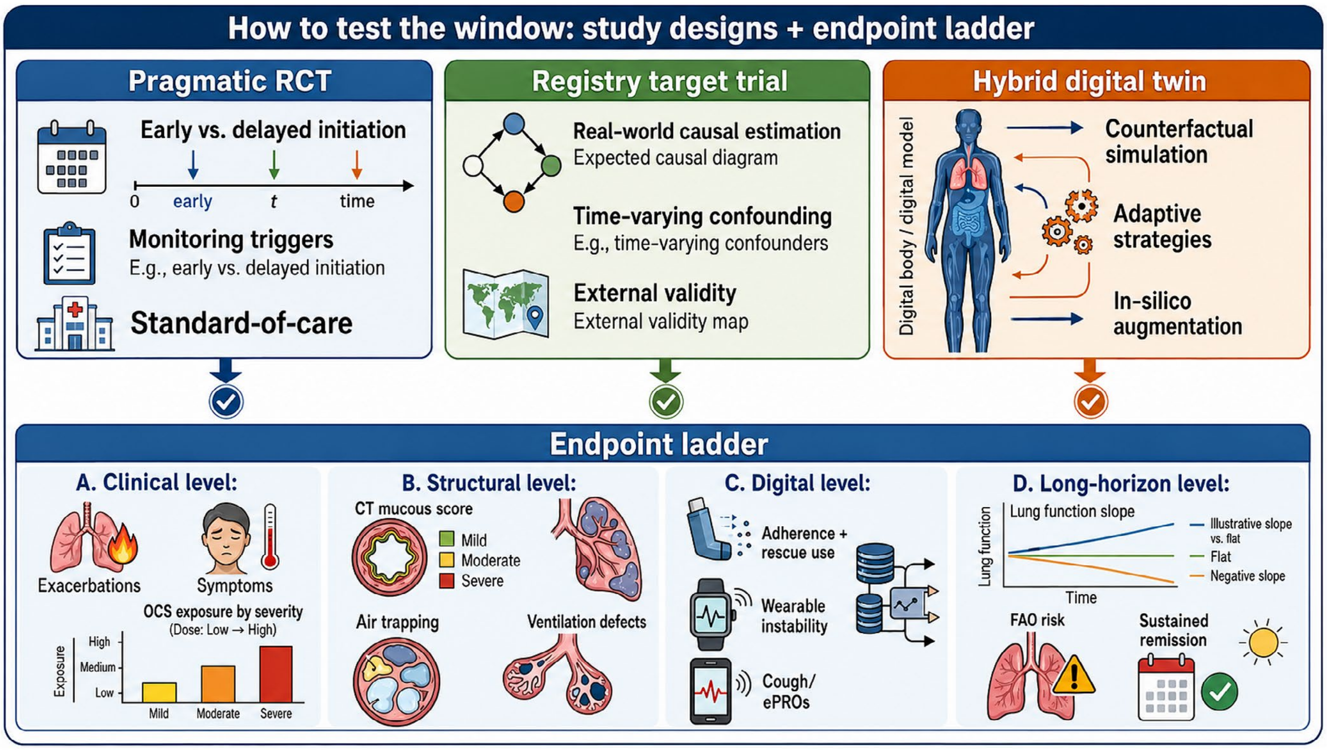

FIGURE 4 | How to test the window of opportunity: Complementary study designs and endpoint hierarchy. Schematic overview of complementary strategies to test the hypothesis that earlier biologic intervention may modify airway remodeling trajectories in severe asthma. The upper panel summarizes three evidence-generation approaches: pragmatic randomized controlled trials (RCTs) comparing early versus delayed biologic initiation within standard-of-care pathways and guided by predefined monitoring triggers; registry-based target trial emulation, aimed at estimating real-world causal effects while addressing time-varying confounding and supporting external validity; and hybrid digital twin approaches, in which multimodal longitudinal data are used for counterfactual simulation, adaptive strategies, and in silico trial augmentation. The lower panel presents a shared endpoint ladder across four levels: Clinical (exacerbations, symptoms, oral corticosteroid exposure), structural (CT mucus score, air trapping, ventilation defects), digital (adherence and rescue use, wearable instability, cough/electronic patient-reported outcomes), and long-horizon outcomes (lung-function slope, fixed airflow obstruction risk, sustained remission). Together, these domains provide a multidimensional framework for evaluating whether th timing of biologic therapy influences both short-term control and long-term structural-functional disease trajectories.

TABLE 1 | Remodeling and timing endpoints: Modality, what it captures, readiness, and recommended use.

Domain	Endpoint/Readout	What it captures	Strengths	Limitations	Readiness (Now/3y/5y)	Recommended use
Biology	Blood eosinophils, FeNO, IgE	T2 activity proxies	Accessible, scalable	Not structural, decoupling possible	Now	Baseline stratification; monitoring
Biology	Breathomics (VOCs/eNose)	"Near-real-time" metabolic/inflammatory signature	Non-invasive, AI-friendly	Standardization/validation needed	3y → 5y	Add-on phenotyping; early signals
Structure	Quantitative CT: mucus plug score	Luminal obstruction + structural-functional trait	Measurable, spatial	Radiation; harmonization	Now→3y	Surrogate structural endpoint
Structure	Quantitative CT: air trapping metrics	Small airways dysfunction	Scalable in cohorts	Protocol variability	3y	Trajectory endpoint
Structure	129Xe MRI ventilation defects	Ventilation heterogeneity	Sensitive, functional	Availability/cost	3y → 5y	Mechanistic + response forecasting
Tissue	Biopsy: RBM thickness/ECM	"Ground truth" remodeling	Mechanistic validity	Invasive, sampling	Now (subset)	Validation/calibration of surrogates
Time	Smart inhaler: adherence + SABA use	Behavior + instability	Objective, continuous	Ecosystem fragmentation	Now	Timing triggers; confounding control
Time	Wearables (HRV/sleep/activity)	Physiologic stress + risk state	Passive, longitudinal	Noise; heterogeneity	Now→3y	Risk windows; relapse detection
Time	Cough monitoring/ePROs	Symptom burden objectively	Sensitive, patient-centered	Validation needed	3y	Add-on instability marker
AI	Causal models/target trial emulation	Timing effect estimates	Strategy-focused	Assumption-sensitive	3y	Window testing
AI	Digital twin + simulation	Counterfactual timing & strategies	Personalized optimization	Requires validation	5y	In silico enrichment

Abbreviations: AI, artificial intelligence; CT, computed tomography; ECM, extracellular matrix; eNOSE, electronic NOSE; ePRO, Electronic patient-reported outcome; FeNO, Fractional Exhaled Nitric Oxide; HRV, Heart Rate Variability; IgE, immunoglobulin E; MRI, magnetic resonance imaging; RBM, Reticular basement membrane; SABA, Short-Acting Beta-Agonist; VOC, volatile organic compounds; y, year.

The first component defines key biological pathways driving airway remodeling. Type 2 cytokine-mediated processes are central, promoting ECM deposition, goblet cell hyperplasia, and persistent inflammation. Epithelial-derived alarmins amplify these cascades, while small airway dysfunction—characterized by ventilation heterogeneity—represents an additional domain. Identifying these pathways anchors decisions in pathobiology rather than symptoms alone [3–8, 10–14, 45–48].

The second component defines a biomarker signature capturing inflammation and disease instability. Traditional markers (eosinophils, FeNO,and IgE) identify type 2 inflammation but fail to reflect dynamic burden. Thus, digital biomarkers—such as rescue use, adherence tracking, cough metrics, and activity or sleep disruption—provide continuous real-world insight, complemented by exposure data. Mechanistic understanding should guide biomarker selection and interpretation [15–20, 84, 85].

The third component focuses on remodeling readouts using longitudinal, scalable endpoints. Imaging modalities such as CT quantify mucus burden and airway structure, while MRI assesses ventilation heterogeneity. In selected settings, biopsy-derived metrics validate structural changes. Integrating imaging and tissue data ensures measurable, time-linked outcomes [9–14, 53–59].

Finally, timing is reframed as a strategy rather than a fixed decision. Approaches may include early biologic initiation in high-risk patients, delayed initiation with intensified monitoring, or adaptive strategies based on longitudinal response. Comparing these strategies enables evaluation of their impact on remodeling, lung function, and long-term outcomes.

Overall, this framework integrates mechanisms, biomarkers, structural readouts, and timing into a clinically applicable model to test whether a therapeutic window exists and how to act on it [24–29, 71, 72].

8 | Future Directions: A Progressive 1-, 3-, and 5-Year Landscape

In the near term—the next 12 months—priority should be given to implementing a minimal yet high-yield "timing dataset" that integrates routine clinical variables with inflammatory biomarkers, objective measures of treatment adherence and rescue medication use, and feasible structural assessments in enriched patient subsets. Embedding objective adherence monitoring and digital endpoints within severe asthma care pathways will be essential to reduce misclassification of poor adherence as refractory biology [18–21, 24, 25]. The systematic adoption of CT-based mucus scoring—and ventilation imaging where available—will help anchor structural phenotypes and enable short-interval tracking of treatment responses that may precede or diverge from changes in conventional lung function measures [13–17, 74, 75, 92, 93]. In parallel, pilot studies on breathomics, built on standardized sampling protocols and predefined external validation strategies, should be introduced as a scalable biological layer aligned with the treatable-traits framework, extending phenotyping beyond established inflammatory markers ([39, 40, 43–45], Figure 3, Figure 4).

Over a medium-term horizon (within 3 years), the development of integrated datasets will support more rigorous causal analyses to unravel the effects of early versus delayed treatment initiation. Target trial emulation and related causal inference methods should be applied to estimate timing-dependent treatment effects under transparent assumptions and control of time-varying confounding [85, 86]. At the same time, imaging-derived measures—particularly mucus burden and ventilation heterogeneity—should be formally evaluated as candidate surrogate endpoints by linking their longitudinal trajectories to clinically meaningful outcomes such as lung-function decline and exacerbation burden [9–11, 15, 16, 74, 75, 94]. Achieving sufficient statistical power and generalizability will require scaling analyses across centers using federated and privacy-preserving infrastructures, enabling collaborative learning without centralized data sharing ([87, 88], Figure 3, Figure 4).

Looking further ahead (within 5 years), advances in machine learning offer the opportunity to move beyond predefined features and labels. Self-supervised learning approaches could be leveraged to model large volumes of unlabeled longitudinal sensor and digital health data, yielding robust representations of disease states and transitions over time [89]. By integrating imaging, conventional and emerging biomarkers—including breathomics—with causal effect estimates, it may become possible to simulate counterfactual timing strategies within patient-specific models. Such models could support exploration of alternative intervention pathways and inform the design of adaptive, timing-sensitive trials aimed at modifying long-term structural and clinical trajectories ([39, 40, 43–45, 85–92], Table 1, Figure 3, Figure 4).

These directions collectively present a cohesive roadmap for a remodeling-aware, timing-centered paradigm in severe asthma. Anchored in quantifiable structural and temporal endpoints and enhanced by scalable biological assessments like breathomics, this approach—supported by causal AI—provides a realistic and credible pathway for progressing beyond symptom control to achieve disease modification [95].

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Consent

A written informed consent was obtained from the parents and informed assent from the patients.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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